6 **Ans: B**

Taking rightward to be positive,

$$5000u_1 + 5000u_2 = 10000(0.50)$$
$$u_1 + u_2 = 1 \qquad \text{--- (1)}$$

and $$\tfrac{1}{2}(5000)u_1^2 + \tfrac{1}{2}(5000)u_2^2 = \tfrac{1}{2}(10000)(0.5^2) + 11250$$
$$u_1^2 + u_2^2 = 5 \qquad \text{--- (2)}$$

Solving the simultaneous equation, we get either

- $u_1 = -1.0$ m s^{-1} and $u_2 = 2.0$ m s^{-1}, OR
- $u_1 = 2.0$ m s^{-1} and $u_2 = -1.0$ m s^{-1}

The first set of answers is rejected because in this case, the carriages are moving away from each other at the start and will never collide.

Hence, only the second set of answers is possible. Therefore, $u_2 = -1.0$ m s^{-1}.

7 **Ans: A**